

Mihir Jayant

Software Engineer · Applied AI · Distributed Systems

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PROFILE

Final-year Computer Science (AI) student at Nottingham Trent University, graduating June 2026 with First Class Honours. Comfortable owning a system from idea to production, and most drawn to the harder parts of the stack: distributed architecture, concurrency, and getting AI features working inside a real product. Recent work includes a six-container distributed load balancer in Java, a three-node Hadoop cluster that processes the full UK road-safety dataset in parallel, and a deployed full-stack auction platform built around a double-entry wallet ledger and a Gemini-powered assistant. Currently a Technical Specialist at Apple, where the daily job is turning a messy technical problem into a clear recommendation someone can act on.

EDUCATION

Nottingham Trent University

Sep 2023 - Jun 2026

BSc (Hons) Computer Science (Artificial Intelligence) · First Class Honours · Clifton Campus, Nottingham

Relevant modules: Artificial Intelligence; Systems Software (distributed load balancing, Docker, concurrency, OS scheduling, networking); Distributed Database Engineering (Hadoop, MapReduce, HDFS, MongoDB, Cassandra); Service-Centric & Cloud Computing (REST, SOA, cloud-native design); Software Design & Implementation (Java, OOP, data structures & algorithms).

TECHNICAL SKILLS

Languages	Java (primary), Python, JavaScript / Node.js, SQL
Applied AI & data	LLM API integration (Gemini); NLP and applied ML (NLTK, TF-IDF, CNN); large-scale data processing with Hadoop / MapReduce
Distributed & infra	Docker & Docker Compose (multi-container orchestration); multi-node Hadoop / HDFS / MapReduce; load balancing (RR, WRR, SJN, FCFS); health checks, semaphore concurrency control, elastic scaling
Backend & APIs	REST API design, microservices, SOA, Apache Tomcat, Node.js / Express, JSON, real-time with Socket.io, ACID transactions
Databases	PostgreSQL, MySQL, SQLite, MongoDB, Azure Cosmos DB, Cassandra; schema design, indexing, parameterised queries
Security	AES encryption, JWT, bcrypt, OAuth, RBAC, OWASP Top 10
Tools & practice	Git, Maven, Linux, Agile / Scrum, automated testing (Jest)

SELECTED PROJECTS

Online Bidding System - Final Year Project

Oct 2025 - Apr 2026

Node.js · Express · PostgreSQL · MongoDB Atlas · React · Socket.io · Gemini · JWT · Render

- Built and deployed a full-stack auction platform to Render: a React front end and a Node/Express API with PostgreSQL and MongoDB Atlas behind it. Live bidding runs over Socket.io, and a Gemini-powered assistant sits inside the product flow.
- The money side runs on an immutable double-entry wallet ledger with ACID transactions in PostgreSQL, so balances always reconcile and no bid can spend the same funds twice.
- Shipped across 8 Agile sprints with a 37-test Jest suite passing at 100% and API responses under 500ms. Auth and access covered by JWT, OAuth, RBAC, and rate limiting.

Load Balancer & Distributed File Transfer System

2024 - 2025

Java · JavaFX · Docker & Docker Compose · MySQL · SQLite · AES · Linux

- Designed a distributed file system across six Docker containers (one load balancer, four storage nodes, and a MySQL metadata server), with health checks that route traffic away from a failed node without dropping in-flight requests.
- Implemented and benchmarked four scheduling algorithms (Round Robin, Weighted Round Robin, Shortest-Job-Next, FCFS) with semaphore-based concurrency control and an aging mechanism to prevent starvation, then measured throughput and fairness under varying load.
- Files split into AES-encrypted chunks across nodes, with local SQLite for session state and remote MySQL for metadata kept consistent via container-level locking. A live console tracks node health and throughput, and new capacity drops in through Docker Compose with no config changes.

Distributed Big Data Analysis - 3-Node Hadoop Cluster

Sep 2025 - Feb 2026

Java · Hadoop · HDFS · MapReduce · Linux

- Set up a three-node Hadoop cluster on Linux VMs from scratch: configured HDFS storage across the nodes, inter-node communication, and health monitoring before submitting any jobs.
- Wrote Java MapReduce jobs to process the full UK Department for Transport road-safety dataset in parallel, aggregating casualty-severity records at a size a single machine can't handle, with output checked against known totals to confirm correctness.

AI Chatbot

Python · NLTK · TF-IDF · CNN

- Built a multi-modal chatbot for the AI module that pairs classic NLP (TF-IDF intent matching) with a CNN, handling both text and image input.

Further: CycleNest, a cloud-hosted Java REST microservice (Apache Tomcat, SOA, MongoDB, deployed on AWS EC2, validated against 10,000+ records); autonomous robot navigation in ROS2 and Docker.

EXPERIENCE

Apple Inc. - Technical Specialist

Jul 2025 - Present

Leicester Highcross

- Diagnose and fix technical issues across Apple's services and devices every day (iCloud sync, Apple ID authentication, MDM device management), usually working through unfamiliar problems under time pressure.
- Translate complicated technical trade-offs into plain recommendations for people who aren't technical, which is the same core skill as working with a product team.
- Finished 2nd in the team for Business Introductions and earned Blue Orchid status, Apple's internal recognition for going beyond the bar on customer satisfaction, productivity, and collaboration.

CERTIFICATIONS & ACTIVITIES

SQL for Data Science, UC Davis (Coursera, May 2026) · Quantum Software Engineering Job Simulation, Forage (Jun 2026) · Student Mentor, NTU (2023-2026) · State-level Badminton Player (India) · Interests: distributed systems, applied AI, JVM performance, gym training.